

# International Migration, Remittance Income, and the Modernization of Rural Areas

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## Abstract

This paper examines international labour migration as a driver of local structural transformation through its effects on remittances flow to migrant-sending economies. Using a shift-share instrumental variable design that combines historical municipality-level migration exposure with destination-country exchange rate fluctuations, we identify that favourable exchange rate shocks increase remittance inflows by up to 21 percent, leading to improved household living standards through asset accumulation and access to modern amenities. These gains coincide with reduced household engagement in agriculture, declining investment in agricultural inputs, and substantial labor reallocation from agriculture to the service sector. Municipalities with higher migration exposure exhibit greater enterprise presence, especially in non-tradable service sector. However, this expansion appears driven primarily by remittance-induced consumption demand rather than productivity improvements. The findings suggest that while remittances enhance household welfare, they do not necessarily boost local productive capacity, underscoring the limits of remittances' long-term transformational role.

**JEL Codes:** F24, O15, J61

**Keywords:** Migration, remittances, structural transformation, exchange rate, labour reallocation

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# 1 Introduction

In 2022, 169 million labor migrants globally remitted USD 831 billion, with 77% of these flows directed to low- and middle-income countries, predominantly in Asia and Africa (IOM, 2024). These remittances constitute a critical source of development finance, directly through enhanced household income and investment capacity, and indirectly via increased state tax revenue. Nepal exemplifies this phenomenon well. Approximately 7.5% of its 30 million population work abroad, generating annual remittances of USD 16 billion (CBS, 2023a). These inflows represent nearly 30% of the country's GDP, substantially surpassing the combined contributions of exports, foreign direct investment, and official development assistance.

Existing literature on international labour migration broadly suggests positive implications of remittances on receiving economies. In particular, remittances represent the most direct financial transfers that can reduce poverty, increase household consumption, and facilitate investments in education and health, and improves subjective well-being (Clemens, Montenegro and Pritchett, 2019; Mobarak, Sharif and Shrestha, 2023; Theoharides, 2018; Yang and Martinez, 2011). Yang and Choi (2007) highlight how migrant remittances respond counter-cyclically to home country shocks, providing crucial insurance against economic volatility. Batista et al. (2025); Khanna and Morales (2025) address the brain drain concerns, like Anelli et al. (2023), by showing that high-skilled emigration from India to the United States stimulated innovation and entrepreneurship in source regions. Beyond remittances, return migration and diaspora networks create additional knowledge transfers and entrepreneurship opportunities (Dustmann, Fadlon and Weiss, 2011).

The primary objective of this paper is to present a nuanced analysis of how migration shapes the nature of local economic transformation in the setting of Nepal. We draw on detailed administrative records covering nearly eight million labour approvals for international migration from Nepal, complemented by the entire population microdata census encompassing 26 million individuals, a census of one million firms, and a panel house-

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The literature also acknowledges potential costs of international migration. For example, selective emigration of skilled workers may constrain institutional development and productivity growth in sending countries (Docquier and Rapoport, 2012).

hold survey. We also construct a shift-share instrumental variable that captures variations in foreign currency exchange rate fluctuations in destination countries. Aggregating all these data to municipality level longitudinal dataset over two decades during the period 2001–2021, our instrumental variable approach finds that remittance windfalls due to favourable exchange rate movement can accelerate the structural transformation of origin local economies, but do not necessarily strengthen their productive capacity, even in the long run. Remittance inflows improve household living standards and access to modern amenities, but also trigger disengagement from agricultural activities, accompanied by decline in the use of key inputs and technology in farming. Consequently, the surplus workforce, particularly unpaid family workers, is reallocated to low-productivity service sectors. Meanwhile, high-migration areas witness a higher concentration of medium to larger enterprises in non-tradable services, most likely spurred by remittance-financed local demand. This pattern suggests demand-driven growth rather than productive transformation of local economies.

The main empirical challenge in our analysis is that migration decisions are endogenous, which often respond to local economic opportunities. Moreover, areas with higher outmigration may systematically differ from those with lower migration in ways that also shape economic outcomes. We address these concerns by taking a shift–share instrumental variable approach, which has become prominent recently following [Borusyak and Hull \(2023\)](#) and [Borusyak, Hull and Jaravel \(2025\)](#). We follow a particular strand of this literature that uses currency exchange rate fluctuations as a source of exogenous variation in remittance flows to analyse how exchange rate fluctuations influence remittances flow and, in turn, affect migrant household welfare and domestic economy ([Hilmy, 2025](#); [Khanna et al., 2025](#); [Yang, 2008](#)). Currency exchange rate fluctuations offer plausibly exogenous variation for identifying the causal effects of remittances, since changes in the value of destination country’s currency affect migrants’ remittances abroad but are unrelated to local economic conditions ([Khanna et al., 2025](#)). Individual migrants have no capacity to influence exchange rates, as their remittance volumes are negligible relative to global currency markets. Migrants’ fixed-term contractual employment arrangements also preclude strategic timing of remittances based on anticipated rate changes. In the case of Nepal,

exogeneity is further reinforced by the institutional setting. The Nepali Rupee is pegged to the Indian Rupee, whose movements against third-country currencies are determined by India’s macroeconomic conditions, and these monetary policy-related factors are entirely external to Nepal’s migration dynamics. Moreover, as a small open economy with no influence on international financial markets, Nepal’s migration flows cannot systematically affect exchange rate movements. Together, these features ensure that fluctuations in the Nepali currency vis-à-vis destination-country currencies constitute plausibly exogenous shocks to the value of remittances received by Nepali households. We discuss the exogeneity of the exchange rate fluctuations further in section 4.1.

Nepal provides an especially compelling setting for examining the developmental consequences of international labour migration for three interrelated reasons. First, Nepal is the ‘basket case of remittance economy’. Roughly one in every two Nepali households has at least one member employed abroad and receives international remittance (CBS, 2023b). This underscores the key role of migration in household welfare and economic development in Nepal. Second, Nepal’s exceptionally rich and comprehensive data enable granular analysis at the local level over a long term. We utilise administrative records covering over eight million labour approvals, the entire population census microdata for 26 million individuals, a detailed census of one-million firms, and a household panel survey, all of which enable us to trace the multidimensional impacts of migration at municipality- and household-level over two decades during the period 2001–2021. Third, Nepali migrants are widely dispersed across 150 destination countries including those in West Asia, Japan, South Korea, Australia, Malaysia, and Europe, which creates substantial cross-destination variation in exposure to exchange rate shocks. This heterogeneity provides strong empirical leverage for identifying the causal effects of currency fluctuations on local economic development.

Our results suggest that remittance windfalls due to foreign currency fluctuations have substantial downstream effects on households decisions, labour market, and firm expansion. First, we show that foreign exchange shocks significantly improve household welfare. When destination-country currencies appreciate against the Nepali rupee, the purchasing power of remittances increases, resulting in improved household living

standards as manifested in transitions to modern amenities, and increased ownership of durable assets. However, these effects are driven primarily by consumption-oriented expenditure rather than productive capital accumulation. Notably, households significantly reduce farming activity, as evidenced by substantial increases in fallow farmland and reduction in key agriculture inputs (like seeds and fertilisers) as well as decreases in capital investments in agricultural equipment. These results suggest that while remittances raise household living standards, they may not lead to enhanced local productive capacity.

Second, foreign exchange shocks trigger substantial labour reallocation. Employment is shifted away from agriculture, particularly among self-employed and unpaid family workers, and toward wage employment in non-farm services. Yet this reallocation moves labour into low-productivity services rather than industrial sectors. The shift is concentrated in elementary and managerial occupations rather than professional roles, which suggests limited potential for productivity gains.

Third, currency shocks to remittances also spur firm growth at the local level, especially among medium and large-sized enterprises. This expansion, however, is confined to traditional service activities such as retail trade, hospitality, with little response in the large industrial sectors. This pattern points to demand-driven growth and a structural shift toward non-tradable services rather than productive diversification of the local economy.

This paper contributes to the literature linking international migration to structural transformation in low-income settings. Earlier work in this area, such as [Dinkelman, Kumchulesi and Mariotti \(2024\)](#); [Khanna et al. \(2025\)](#); [Mobarak, Sharif and Shrestha \(2023\)](#); [Yang \(2008\)](#), highlights how international migration can initiate structural transformation through remittance-financed investment in rural non-farm enterprises. Our findings nuance this narrative by showing that that remittance windfalls substantially enhance household welfare and living standards but do not automatically translate into productive capital accumulation or sectoral upgrading. Our paper therefore paints an atypical picture of structural transformation that does not follow the usual growth path from agriculture to industry to services. Instead, remittance windfalls may prompt local economies to skip

the industrial phase altogether (??). This pattern echoes broader evidence that remittances alone are insufficient to trigger sustained structural change without complementary policy and institutional support (Clemens and Postel, 2019).

Moreover, our research adds to a related strand of literature that emphasises labour reallocation as a key mechanism of transformation. Dinkelman and Mariotti (2017); Dinkelman, Kumchulesi and Mariotti (2024) and Porzio, Rossi and Santangelo (2022) demonstrate that migration can shift labour out of agriculture, often through skill accumulation and non-farm opportunities, while Ngai and Petrongolo (2017) show how rising services reshape gendered labour outcomes. Our findings qualify these insights by showing that employment declines sharply in agriculture, particularly among self-employed and unpaid family workers, while increasing in non-farm employment. However, this growth is concentrated in low-productivity service and elementary occupations, with modest gains among managers but none among professionals. This pattern represents a service-oriented reallocation with limited potential for productivity growth necessary for a sustained transformation.

Our further contribution to the literature on structural transformation is to link remittances, migration, and firm-level responses. The expansion of enterprises in response to rising local demand aligns with Yang (2008), who shows that migration-driven income shocks can stimulate domestic activity. Like other studies, we find that expansion often remains confined to non-tradable sectors rather than manufacturing or tradable industries (??). In our case, firms' growth is likely driven by remittance-fueled consumption rather than productive investment, representing "remittance-led demand" rather than "investment-led transformation" (Bustos, Caprettini and Ponticelli, 2016; Imbert, Papp and Ramos, 2022).

The rest of the paper is organised as follows: Section 2 provides contextual information on international labour migration from Nepal as well as a brief description of its economic structure. Section 3 describes the data sources and Section 4 discusses in detail the empirical approach taken in the paper. Section 5 presents the results. Finally, Section 6 concludes.

## 2 Contextual Background

### 2.1 International Labour Migration from Nepal

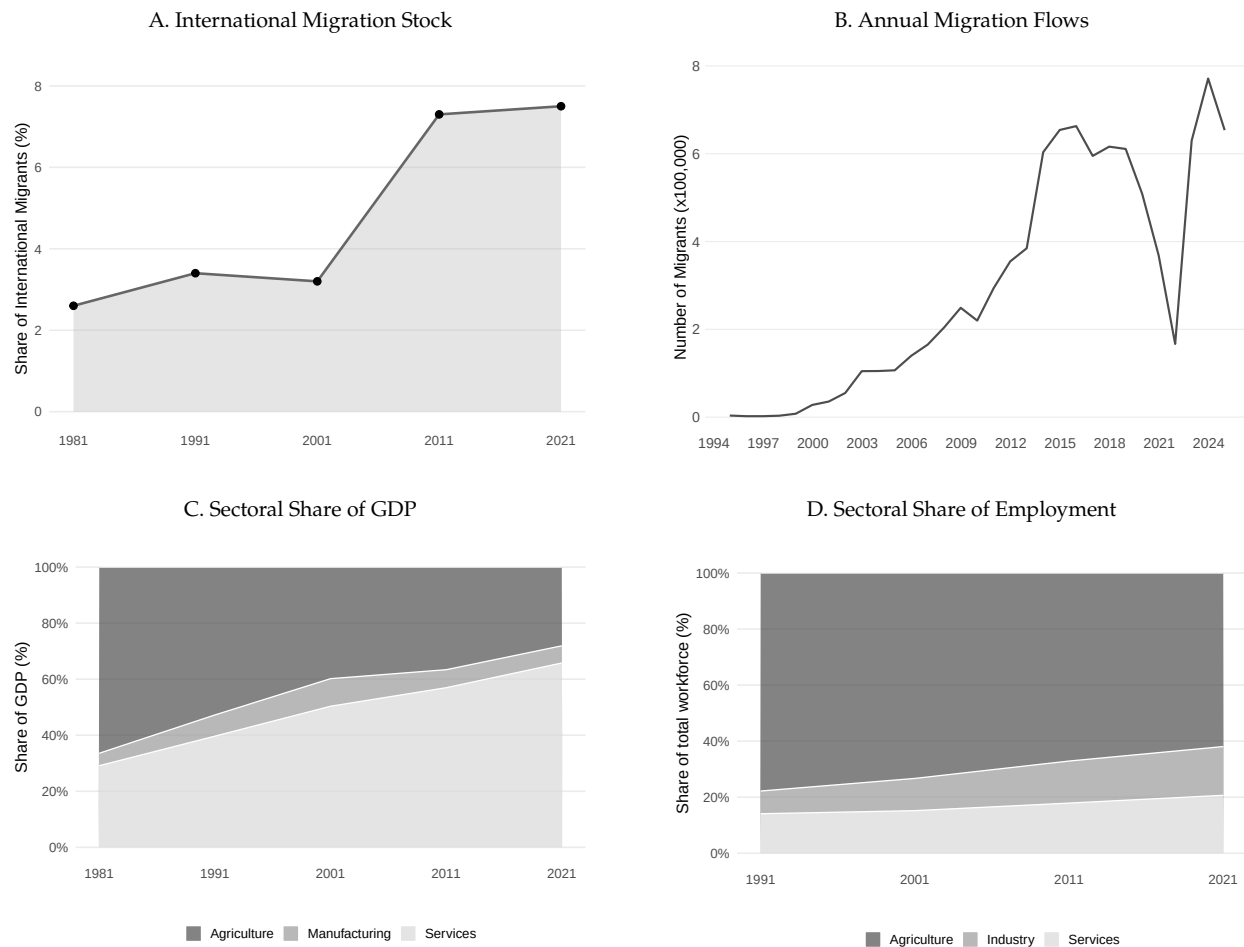
International labour migration has been a defining feature of Nepal's economy and labour market. The trend of migration has rapidly increased over the past two decades (see Fig. 1A). The number of Nepali citizens working overseas has trippled from 0.7 million in 2001 to approximately 2.2 million in 2021 (CBS, 2023a). Administrative records further highlight the massive scale of labour migration. Figure 1D shows annual labour migration flows of around 700,000 workers in recent years. Cumulatively, nearly eight million labour permits for foreign employment have been issued between 2009 and 2024 across over 150 destinations worldwide.

Nepal's sluggish economic performance has constrained the development of productivity-enhancing and well-paying labour market, particularly for its predominantly young population. As a result, international labour migration has become a central livelihood strategy. Census records indicate that 72% of Nepalis migrating for work are between the ages of 15 and 35. Most have limited education: 41% completed only basic schooling and 45% up to secondary level. (CBS, 2023a). Accordingly, Nepali migrants tend to be absorbed into low-skilled occupations such as general labourers, construction workers, cooks, and drivers in destination countries (Fig. A1). International migration from Nepal is overwhelmingly male-dominated; four out of five emigrants are men. Migrants are typically drawn from rural areas where agriculture remains the primary source of livelihood (CBS, 2023a).

There are also pronounced regional disparities in migration patterns which reflects differences in local economic conditions, migration networks, and historical trajectories of labour mobility. For instance, as of 2021, the Arghakhanchi district had approximately one in every three adult men working abroad, whereas the Dolpa district recorded only around one migrant per hundred adult men. These contrasts highlight how migration is deeply embedded in the social and economic fabric of some districts, while remaining limited in others.

Historically, India was the primary destination for Nepali people seeking employment opportunities abroad. Countries in Western and South-Eastern Asia have emerged as the

Figure 1: Trends in international migration and structural change in Nepal



Notes: Panel A depicts the absentee population due to out migration, as recorded in the decennial census. Panel B shows annual international labour migration outflows during the period 1995 to 2024. Panel C presents the value added of Agriculture, Industry, and Services sectors as a share of GDP, whereas, Panel D displays the distribution of the workforce across the three sectors.

principal destinations for Nepali labour migrants in the past two decades. The top six destinations including Qatar, the United Arab Emirates, South Korea, Saudi Arabia, Japan, and Malaysia account for the majority of outflows. In recent years, European destinations such as Romania, Croatia, Malta, and Poland have also attracted substantial numbers of job-seekers, despite Europe's more restrictive immigration regimes. The Government of Nepal has signed bilateral labour agreements with several destination countries to promote safe foreign employment, starting with Qatar in 2005. These agreements typically

The other bilateral labour agreements Nepal has concluded are with Republic of Korea (2007, Employment Permit System), Bahrain (2008, labour and training), Japan (2009, technical interns; 2019, specified skilled workers), Jordan (2017, manpower), Malaysia (2018, recruitment and repatriation), Mauritius (2019, recruitment and employment), Israel (2020, agriculture and caregiving), and the United Kingdom (2022, healthcare professionals).



cover critical aspects of worker protection including minimum wage guarantees, decent work conditions, and fundamental worker rights. The agreements aim to facilitate orderly migration flows by establishing formal channels for recruitment and placement, while simultaneously providing legal frameworks for worker protection against exploitation and abuse.

Domestically, foreign labour migration is governed by a well-defined legal and administrative framework. All prospective migrants must obtain labour permits before departure, and compliance is enforced through dedicated immigration desks at Tribhuvan International Airport, which is the primary exit point for migrant workers. The migration process is primarily facilitated by licensed recruitment agencies that serve as intermediaries between foreign employers and Nepali workers. These agencies secure employment contracts from labour-importing countries and recruit candidates nationwide, managing documentation, permit processing, visa coordination, pre-departure training, and travel logistics. While this regulatory architecture is designed to formalise migration and reduce risks against labour exploitation, it also reflects the heavy reliance on private recruitment intermediaries.

Nepali migrant workers are typically employed on fixed-term contracts lasting two to three years. Key employment terms are finalised before departure through mandatory contract signing and compulsory pre-departure orientation training. These contracts stipulate wages in the destination-country currency, which usually remain fixed over the contract period and provide a degree of income certainty. Upon completion of their contracts, migrants return to Nepal but frequently re-enter foreign employment, either by renewing contracts in the same destination or by shifting to alternative labour markets. Most migrants repeat this migration cycle two to three times over their working lives, creating a pattern of circular and temporary labour mobility. This system enables households to smooth income over time, accumulate savings, and finance major expenditures, while allowing migrants to maintain strong social and economic ties to their home communities.

## 2.2 Structure of the Nepali Economy

Nepal's economy remains heavily reliant on agricultural employment despite the sector's declining value added in GDP. As Figure 1C shows, the share of agriculture in GDP has declined steadily over the recent decades (22% in 2023), while growth in the industrial sector has remained largely stagnant. By contrast, the service sector has expanded rapidly and accounts for more than 65% of total output in 2023. However, much of this growth is concentrated in low-productivity informal activities such as retail trade, transport, and hospitality, rather than in high-value-added services.

Figure 1D shows that the sectoral reallocation of the workforce has progressed much more slowly. Despite agriculture's declining share in GDP, the sector still employs around 67% of the economically active population, while industrial sectors have failed to generate sufficient new jobs. The shortage of productive employment opportunities in non-agricultural sectors coupled with limited access to capital and skills training has kept much of the workforce reliant on subsistence farming. Consequently, many young people have sought foreign employment as an alternative means of livelihood.

This pattern of structural transformation differs from the historical path of now-advanced economies. In the classical development pattern, countries moved from agriculture to manufacturing and then to high-productivity services. This sequence of development supported job creation, productivity growth, and technological progress as industry absorbed rural labour and enabled capital accumulation (Gollin and Kaboski, 2023; ?). In Nepal, however, the shift out of agriculture has occurred without a corresponding expansion of industry. Economic activity has moved directly into low-productivity informal services which resulted in a scarcity of quality jobs and weak productivity growth.

These structural patterns are mirrored in the enterprise sector which is dominated by small and medium-sized enterprises. These firms are typically family-owned, labour-intensive, and concentrated in wholesale and retail trade, small manufacturing, and basic non-tradeable services. Most firms operate informally, which limits their access to finance, integration into markets, and technology transfer. The limited number of large enterprises that exist are predominantly involved in imports.

### 3 Data

**Population Census Data:** Our primary source of migration data is the Population and Housing Census of Nepal for the years 2001, 2011, and 2021. The 2001 and 2011 rounds provide record of the entire population, covering approximately 22 million and 26 million individuals, respectively. We also use a representative sample of 3.2 million individuals from the 2021 census, encompassing 7.5% of the total population. All census data are collected by the Central Bureau of Statistics of Nepal. These datasets contain detailed individual-level information on migration, educational attainment, demographic attributes, occupation, and employment status. They also include household-level indicators such as housing characteristics, amenities, and asset ownership.

**Administrative Migration Data:** We supplement the census data with administrative migration records from the Department of Foreign Employment, which maintains official documentation of foreign labour employment approvals. This dataset contains approximately eight million individual migration records for 2009-2024 and includes detailed information on departure dates, origin districts and municipalities, destination countries, occupations, and skill categories. We do not have access to migrants' contracted wages or salaries. Using all the available data, we construct a destination-country-year panel covering the period 2009-2024, which enables us to test the migration response to exchange rate fluctuations.

**Households Panel Survey Data:** We also draw on the Nepal Household Risk and Vulnerability Survey (HRVS) produced by the World Bank ([World Bank, 2020](#)). This nationally representative panel survey tracks approximately 6,000 households across three waves in 2016, 2017, and 2018. HRVS collects rich socioeconomic information, including household demographics, income and expenditure, economic activities, asset holdings, and exposure to various shocks. Importantly, the survey provides detailed information on migration, remittance receipts, financial behaviour, and investment decisions.

**Enterprise Data:** We also utilise the National Economic Census 2018 that covers the universe of non-agricultural and registered agricultural firms across Nepal. The census

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The Bureau has been renamed as National Statistical Office in 2023.

enumerates 923,356 establishments, and records their registration status, legal form, operational characteristics, record-keeping practices, infrastructure access, and employment details. Firms are classified by economic activity, size, year of establishment, and geographical location. The census also contains detailed financial information on revenue, expenditure, capital investment and credit access.

**Exchange Rate Data:** We source currency exchange rates from a dataset maintained by the Statistics Department of the International Monetary Fund. The dataset provides historical data on bilateral exchange rates between the US Dollar and other national currencies, including both period-average and end-of-period rates. We compute the exchange rate between each Nepali migrant destination country’s currency and the Nepali Rupee using the cross-rate method.

## 4 Empirical Methodology

Estimating the causal effect of international migration on local economic development of migrants’ origin places presents substantial identification challenges due to multiple sources of endogeneity. For example, while remittances from abroad may stimulate local consumption and investment, areas experiencing sluggish economic growth or limited employment opportunities, may simultaneously experience higher emigration rates. Furthermore, areas with strong migrant networks, greater human capital, and superior connectivity may experience both higher international migration propensities and better economic outcomes through channels unrelated to remittances. Therefore, Ordinary Least Squares estimation alone is unlikely to isolate the direct effect of migration from these other factors and thus cannot identify the true causal relationship.

Following [Borusyak and Hull \(2023\)](#) and [Borusyak, Hull and Jaravel \(2025\)](#), we address these concerns by using the shift-share instrument variable approach with an exogenous shift variable. This identification approach combines two key components to construct the instrument. The “shift” comprises country-specific exchange rate fluctuations, which is a proxy for exogenous changes in labour demand conditions at destination countries. These destination-country currency shocks are plausibly exogenous to conditions

within individual municipalities and operate independently of local economic circumstances. We discuss the exogeneity of exchange shocks in the next section. The “share” comprises each municipality’s historical migration exposure to different destination countries. The composition of migrant destinations in 2001 (the baseline) is predetermined by past migration decisions, social networks, and geographical proximity to migration routes rather than by current economic conditions in the municipality of origin. By leveraging differential exposure to currency fluctuations based on pre-existing migration networks, instruments generate plausibly exogenous variation in currency shocks across municipalities that are uncorrelated with unobserved municipal characteristics, thereby enabling causal identification.

#### 4.1 Exogeneity of Currency Exchange Shock

Our analysis considers exchange rate movements as a source of exogenous variation. Bilateral exchange rate fluctuations directly affect the value of remittances in Nepal and shape migration incentives. For example, an appreciation of the destination-country currency relative to the Nepali rupee increases the real value of remittances, raising the returns to migration and encouraging additional outflows. Conversely, depreciation reduces the purchasing power of remittances, potentially limiting migration to that destination. Because foreign exchange shocks are destination-specific and plausibly orthogonal to household migration decisions, they provide suitable exogenous variation for identifying the causal effect of economic incentives on international labour flows within the shift–share framework. We construct the currency exchange rate (FXRate) between destination-country to Nepali Rupees (NPR) relative to 2001, which is the baseline year of our analysis, as follows:

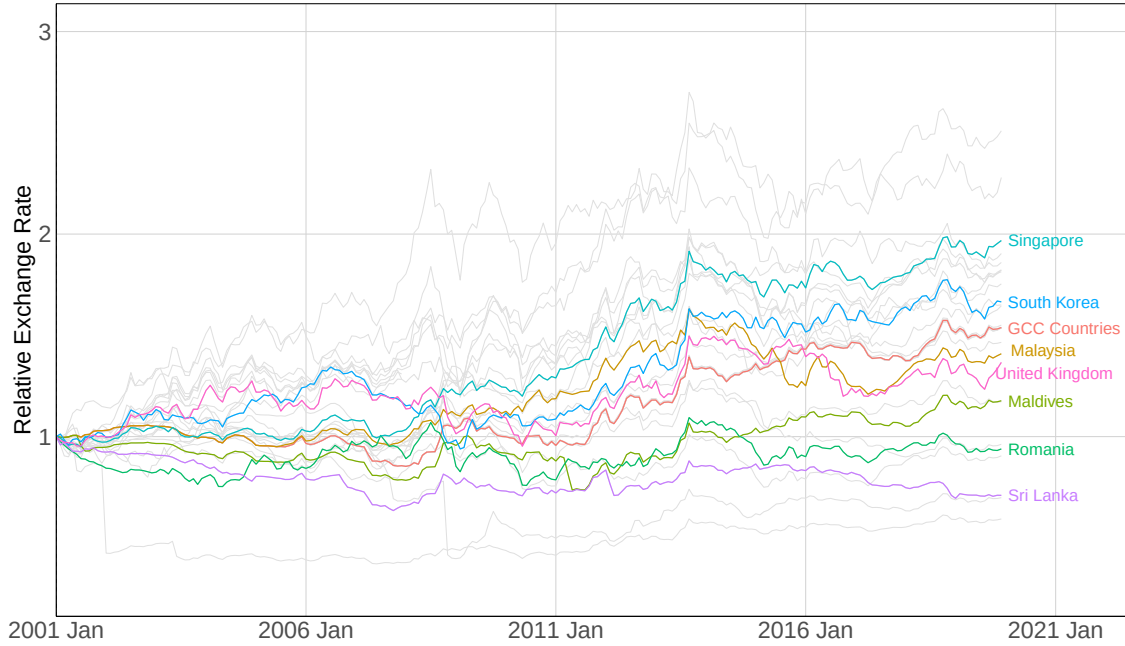
$$\text{FXRate}_{it} = \frac{\text{FX rate to NPR}_{ct}}{\text{FX rate to NPR}_{c,2001}} \quad (1)$$

Figure 2 shows the movement of relative currency shocks over the period 2001–2021.

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We note that the exchange rate of each destination-country currency against the Nepali rupee (NPR) is not directly available in the data, as all rates are reported relative to the US dollar (USD). Therefore, we derive the destination-country-to-NPR exchange rate as a cross rate to USD.

Figure 2: Destination-country-currency-to-NPR relative exchange rate



Notes: The figure shows destination countries' currency exchange rates against the Nepali rupee, relative to January 2001. Coloured lines highlight relative foreign exchange rate fluctuations for selected migrant destination countries.

The *FXRate* variable exhibits variation across major migrant destination countries, and the extent of municipal-level exposure to these shocks varies accordingly.

A key assumption for the causal interpretation of the effects of the *FXRate* is that exchange rate fluctuations are exogenous to Nepal's migration dynamics and household remittance behaviour. We argue that this assumption is plausible because individual migrants do not influence exchange rate movements. Their remittance transfers are negligible relative to international currency markets and cannot determine exchange rate changes. Moreover, migrants are unlikely to determine their migration decisions in anticipation of future exchange rate fluctuations, given both limited access to financial forecasting information and the contractual nature of foreign employment, which constrains such behavioural responses at least in the short run.

The exogeneity of foreign exchange variation is further supported by Nepal's institutional and macroeconomic features. First, the Nepali rupee (NPR) is pegged to the Indian rupee (INR), and India accounts for approximately 72% of Nepal's total trade (?). As a

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We also note that China is Nepal's second-largest trading partner, accounting for approximately 12% of Nepal's trade volume.

result, NPR movements against the third-country currencies largely mirror INR fluctuations, which are in turn driven by India’s macroeconomic conditions, monetary policy, and bilateral exchange rate dynamics with major economies. These factors are entirely external to Nepal and unrelated to local labour market dynamics. Second, Nepal is a small open economy with no meaningful influence on global currency markets, making it implausible that migration flows to major destination countries such as Malaysia, Saudi Arabia, Qatar, or the UAE could exert any systematic feedback on exchange rate movements. Overall, these institutional and market characteristics strengthen the case for treating bilateral currency shocks faced by Nepali migrants as plausibly exogenous.

## 4.2 Do Migration and Remittance Flows Respond to Currency Shocks?

This section examines the first step of the shift-share instrumental variable approach: how destination-specific migration flows and remittance transfers to origin households respond to relative foreign exchange shocks. The theoretical premise underlying this relationship is that an appreciation of a destination country’s currency relative to the migrants’ home currency increases the value of earnings when converted into domestic purchasing power. This increase in the economic returns to migration makes that destination more attractive to prospective migrants and also boost additional remittance inflows into the origin areas. Given that remittance transmission constitutes a primary objective for labour migrants, exchange rate dynamics play a pivotal role in migrants’ destination choice. To empirically test this hypothesis, we estimate the relationship between *FXRate* computed as per equation (1) and aggregate migration outflows as well as household-level remittance receipts.

**Migration outflows:** We utilise comprehensive data on annual migration outflows, which span the period 2009-2024. During this window, approximately eight million individuals obtained labour approvals for foreign employment, which we aggregate to country-year level. We then estimate annual labour migration to destination country on exchange rate shock as below:

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However, China hosts almost no Nepali labour migrants.

The number of migrants may include repeat migrants who require new permits for each deployment and can select alternative destinations upon re-application.

$$\text{Migrants}_{it} = \alpha + \beta \text{FXRate}_{i,t-1} + \theta_t + \phi_i + \epsilon_{it} \quad (2)$$

$\text{Migrants}_{it}$  represents the number of migrants to destination  $i$  in year  $t$ .  $\text{FXRate}_{i,t-1}$  captures the foreign exchange rate shock, which is lagged by one period to mitigate the concerns of contemporaneity.  $\theta_t$  denotes the year-fixed effects, and  $\phi_i$  denotes the country-fixed effects.

**Household remittance receipts:** We also estimate the effect of foreign exchange shock on household remittance receipts using a three-year (2016-2018) panel dataset comprising nearly 6,000 households, of which 2,566 households received at least one remittance transfer during the survey period. The inclusion of household fixed effects helps mitigate self-selection bias that often arises in cross-sectional analyses of remittance outcomes. By exploiting the within-household variation over time, this approach controls for unobserved, time-invariant characteristics such as preferences, social networks, or initial wealth of the migrant households, which may simultaneously influence both the decision to migrate and the amount of remittances received. The estimation equation is specified as below:

$$\text{Remittance}_{it} = \alpha + \beta \text{FXRate}_{i,t-1} + X_{it}\gamma + \theta_t + \phi_i + \epsilon_{it} \quad (3)$$

$\text{Remittance}_{it}$  denotes the amount of remittance received by household  $i$  in year  $t$ ,  $\text{FXRate}_{i,t-1}$  represents the lagged foreign exchange shock,  $X_{it}$  includes time-varying household characteristics,  $\theta_t$  denotes the year-fixed effects which absorb the changes common to all respondents in each survey year, and  $\phi_i$  denotes household fixed effects that absorb the migrant households' invariant characteristics. The control variables include the duration (in weeks) that migrants have remained in the destination country and the frequency of remittance flows during the sample period. Standard errors are clustered at the household level.

Table 1 presents the estimation results for equation (2) and equation (3) in the top and bottom panels, respectively. The top panel shows that a one-unit increase in the

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The government records remittance inflows only at the aggregate level. Banks and financial institutions through which migrants send remittances likely maintain individual-level records, but these are not publicly available.



Table 1: Effect of *FXRate* on migration flows and remittance receipts

|                      | External Migration Flows (Mean = 3,033.5) |                            |                          |                            |
|----------------------|-------------------------------------------|----------------------------|--------------------------|----------------------------|
|                      | (1)                                       | (2)                        | (3)                      | (4)                        |
| FXRate               | 2,517.0**<br>(1,254)                      | 2,289.8*<br>(1,189)        | 4,862.4**<br>(2,175)     | 3,455.3**<br>(1,623)       |
| Country FE           | No                                        | No                         | Yes                      | Yes                        |
| Year FE              | No                                        | Yes                        | No                       | Yes                        |
| Observations         | 1,854                                     | 1,854                      | 1,854                    | 1,854                      |
| R <sup>2</sup>       | 0.004                                     | 0.009                      | 0.804                    | 0.808                      |
|                      | Remittance Receipts (Mean = NPR 193,724)  |                            |                          |                            |
|                      | (1)                                       | (2)                        | (3)                      | (4)                        |
| FXRate               | 23,363.1*<br>(10,879.8)                   | 26,657.9***<br>(8,406.4)   | 23,604.5*<br>(11,057.8)  | 26,730.4***<br>(8,337.8)   |
| Remittance Frequency |                                           | 106,820.4***<br>(20,154.6) |                          | 105,421.7***<br>(19,662.2) |
| Migration Duration   |                                           |                            | 41,048.7**<br>(13,089.2) | 19,662.4*<br>(8,264.5)     |
| Year FE              | Yes                                       | Yes                        | Yes                      | Yes                        |
| Household FE         | Yes                                       | Yes                        | Yes                      | Yes                        |
| Observations         | 4,184                                     | 4,184                      | 4,184                    | 4,184                      |
| R <sup>2</sup>       | 0.463                                     | 0.594                      | 0.469                    | 0.596                      |

Notes: Panel A: The unit of analysis is country-year between 2009-2024. The panel reports estimates of external migration flows with country and year fixed effects, where the standard errors are clustered at the destination-country level. Panel B: The unit of analysis is households-survey years of 2016, 2017 and 2018. The panel reports the estimates of the amount of remittances received with year and household fixed effects, where the standard errors are clustered at the destination country level. The *FXRate* represents relative exchange rate computed as per equation (1). Significance: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

destination-specific currency exchange shock (relative to 2001) is associated with up to 3,500 additional migrants leaving for that destination each year, suggesting that favourable exchange rate movements strengthen migration incentives. The bottom panel demonstrates that a one-unit appreciation of the host-country currency relative to the Nepali rupee since 2001 increases annual remittances for migrant households in Nepal by approximately 12–14% above the mean remittance level of NPR 194,000. This positive relationship between remittances and exchange rate shocks remains robust after accounting for key controls that might influence remittance sending decision, including remittancy frequency and migration duration. Overall, these findings support the theoretical expectation that favourable exchange rate movements increase migration flows and remittance receipts to households in origin.

A critical alternative explanation for our findings is that migrants may simply switch

their remittance transmission methods from informal to formal channels in response to exchange rate shocks, rather than altering their overall remittance behavior (Clemens, 2018). Under this channel substitution hypothesis, the observed changes in formally recorded remittances would represent a spurious correlation rather than a genuine response to foreign exchange market conditions, as the total volume of remittances (formal plus informal) might remain constant while only the composition across channels changes.

However, several factors suggest that channel switching is unlikely to fully explain our results. First, while informal channels were historically dominant, formal remittance channels have significantly closed the gap in recent years (Beck and Peria, 2011). Moreover, the cost of sending remittance to Nepal is below global average, reducing the incentives for migrants to choose the informal channel (MOLESS, 2022). Further, descriptive analysis shows that the vast majority of migrants remit through formal channels, primarily banks and money transfer operators. Figure A2 shows that formal channels capture 98% of all remittance transactions, while informal channels such as *hundi* or cash carrying account for less than two percent, indicating negligible potential for switching effects. This finding aligns with broader trends toward formalisation of remittance markets, driven by regulatory improvements, technological innovations, and enhanced financial infrastructure in both sending and receiving countries (Beck and Peria, 2011).

### 4.3 Construction of the Shift-Share Instrument

The previous section discussed the exogeneity of exchange rate fluctuations as shift component of the shift-share instrument. This section introduces the share component that captures historical migration exposure of each municipality,  $w_{mc}$ :

$$w_{mc} = \frac{migrants_{mc}^{2001}}{\sum_c migrants_{mc}^{2001}} \quad (4)$$

where  $migrants_{mc}^{t=2001}$  denotes the number of migrants from municipality  $m$  that migrated to destination country  $c$  in the baseline year, 2001. We construct migration exposure shares based on the 2001 migration patterns to ensure that destination choices are not contemporaneously influenced by foreign exchange shocks.

Figure A3 shows migration exposure at the municipality level, capturing spatial heterogeneity in historical (2001) migration patterns across key destination countries. This variation provides the basis for identifying how subsequent destination-specific exchange rate shocks differentially affected the locality of origin. Because municipalities send migrants to different destinations in varying proportions, they experience differential exposure to foreign exchange movements of that country’s currency (expressed in Nepali Rupees). The shift-share instruments exploit this heterogeneity to trace how additional remittances induced by currency fluctuations affect local economies.

We further highlight this mechanism by contrasting the migration exposure of two municipalities—Tribeni and Patan—to illustrate the logic of our identification strategy. Tribeni has roughly 27% of its migrants working in Saudi Arabia, compared to only four percent from Patan working in that country. As a result, Tribeni is substantially more sensitive to fluctuations in the Saudi Riyal: an appreciation of the Riyal would translate into larger remittance gains per migrant for Tribeni than for Patan. Conversely, Patan has a much higher concentration of migrants in Malaysia (46%) compared to just eight percent from Tribeni, which makes Patan more exposed to movements in the Malaysian Ringgit. This heterogeneity in destination-country composition generates systematic variation in municipality-level exposure to foreign exchange shocks, which is independent of total migration intensity and therefore credible for identifying differential remittance effects across local economies. Moreover, the instruments can identify the effect of currency fluctuations even when baseline migration patterns may be endogenously determined (Borusyak and Hull, 2023; Borusyak, Hull and Jaravel, 2025).

We operationalise this conceptual framework by combining two key components discussed so far: the exposure share ( $w_{mc}$ ), which represents the municipality’s share of migrant population across destination countries in 2001, and the shift component,  $FXRate_{mt}$ , which captures destination-country currency exchange rate fluctuations relative to the 2001 baseline average. The shift-share instrument is formally constructed as:

$$FXShock_{mt} = \sum_c (w_{mc} \cdot FXRate_{ct}) \quad (5)$$

where the summation across all destination countries  $c$  ensures that aggregate foreign exchange exposure of each municipality reflects the weighted average of currency movements in all countries where its migrants are employed. This set-up also ensures that municipalities with larger migrant shares in countries experiencing stronger currency movements receive larger instrumental values. Figure A4 shows the municipality-level variation in the shift-share instrument.

Beside the composition of migrants across destination countries, another key factor shaping a municipality's exposure to foreign exchange shocks is the overall migration rate (Hilmy, 2025; ?). Two municipalities may send the same share of migrants to the same set of destinations, yet differ substantially in the scale of migration. For example, one municipality may have six percent of its population abroad while another has only one percent. In such cases, the municipality with the larger migrant share is more exposed to exchange rate fluctuations and, as a result, is more likely to experience stronger effects on local economic outcomes. To capture this differential exposure, we weight the foreign exchange shock by baseline migration intensity at the municipality level, measured as the share of migrants in the total municipal population. Formally, migration intensity is defined as the natural log of migrant workers per million population in 2001.

$$\text{MigIntensity}_m^{2001} = \log\left(\frac{\text{total migrants}_m}{\text{population}_m}\right). \quad (6)$$

As with migration shares, we use the 2001 migrant population to avoid contemporaneous migration responses. This weighting implies that foreign exchange shocks in destinations hosting larger Nepali migrant populations carry greater weight in identifying variation.

#### 4.4 Estimating Equation

Favourable exchange rate movements can amplify economic impacts in origin areas by increasing the purchasing power of remittance inflows, and vice versa. This mechanism generates effects across recipient households, labour markets, and firms in the municipality of origin. We estimate the effect of  $FXShock$  on these local economic outcomes using

the following specification:

$$\begin{aligned}
Y_{mt} = & \beta_1 \text{FXShock}_{mt-1} + \beta_2 (\text{FXShock}_{mt-1} \times \text{MigIntensity}_m^{2001}) + \beta_3 (\text{FXShock}_{mt-1} \times \text{Post}) \\
& + \beta_4 (\text{MigIntensity}_m^{2001} \times \text{Post}) + \beta_5 (\text{FXShock}_{mt-1} \times \text{MigIntensity}_m^{2001} \times \text{Post}) \\
& + X'_{mt} \delta + \alpha_m + \gamma_t + \epsilon_{mt}
\end{aligned} \tag{7}$$

$Y_{mt}$  denotes the outcome of interest for municipality  $m$  in census year  $t$ . We examine the effects across three interconnected dimensions: household-level economic behavior, sectoral employment reallocation, and firm-level outcomes. Our coefficient of interest is  $\beta_5$ .

The key explanatory variable  $\text{FXShock}_{mt-1}$  captures the shift-share instrumental variable lagged by one period to avoid contemporaneity concerns.  $\text{MigIntensity}_m^{2001}$  is the share of international migrants in the total population of municipality  $m$  in 2001.  $\text{Post}_t$  is a binary indicator equaling zero for the base year and one for subsequent years.  $X'_{mt}$  includes time-varying municipality controls, including infrastructure access and demographic variables. The specification incorporates the municipality fixed effects  $\alpha_m$  and the year fixed effects  $\gamma_t$  to control for time-invariant characteristics and common temporal shocks, respectively. Standard errors are clustered at the municipality level.  $\epsilon_{mt}$  denotes the error term.

We estimate equation (7) across different time period and units of analysis. For long-term outcomes at the municipality level, we use census years  $t \in \{2001, 2011, 2021\}$ . We analyse the short-term household-level outcomes for  $t \in \{2016, 2017, 2018\}$ . All household-level estimations include the household fixed effects alongside the year fixed effects to absorb time-invariant household characteristics and common temporal shocks. Firm-level effects are estimated using a version of equation (7) for  $t = 2018$ , that is, without any time dimension, given the availability of only cross-sectional data.

## 5 Results and Discussion

### 5.1 Household Investment Decisions

This section discusses effects of *FXShock* on household investment decisions. Households may choose to use additional remittance income to purchase durable goods or improve household amenities. They may also invest in productivity-enhancing assets that increase their future earning potential. For instance, agricultural households may invest remittances in improved technologies like modern irrigation systems, quality seeds, or mechanised equipment that would enable more productive farming. Table 2 reports the estimation results across three outcome categories: household assets, household amenities, and agricultural practices.

Panel A documents an increase in household ownership of modern durable goods, like refrigerators and motorcycles, in municipalities that are exposed to higher *FXShock*. Ownership of high-value assets such as cars are more than doubled, compared to baseline levels. Meanwhile, the share of households with no ownership of consumer durables falls by about 40%, and reliance on low-value assets such as radios or bicycles also declines substantially. These patterns demonstrate that households use remittance income to increase household assets.

Panel B reports that exposure to favourable foreign exchange shocks results in substantial improvements in household living conditions. A large share of households shifts from traditional fuels to modern energy sources, with wood and kerosene use declining sharply while LPG adoption rising by more than 130%. The share of households with access to improved sanitation like modern toilet facilities increases by 30%. However, piped water access does not change, which may suggest that remittance gains translate more readily into private household investments.

Panel C displays that foreign exchange shocks significantly affect some household agricultural practices in the dry season. Households in high-migration municipalities reduce self-cultivation by up to four percent relative to the baseline mean. However, there is no corresponding expansion of land rental markets, as neither renting out nor sharecropping increases. Instead, the likelihood of households leaving land fallow rises by as much

Table 2: Effect of *FXShock* on household assets, amenities, and farming

| <b>Panel A: Household Assets</b>            |                      |                       |                        |                      |                      |                        |                    |                       |
|---------------------------------------------|----------------------|-----------------------|------------------------|----------------------|----------------------|------------------------|--------------------|-----------------------|
|                                             | Radio                | Television            | Bicycle                | Motorcycle           | Car                  | Refrigerator           | Phone              | No Durable            |
| FXShock $\times$ MigIntensity $\times$ Post | -0.180***<br>(0.048) | -0.004<br>(0.030)     | -0.064**<br>(0.038)    | 0.030<br>(0.026)     | 0.013***<br>(0.004)  | 0.055**<br>(0.026)     | 0.034<br>(0.012)   | -0.088***<br>(0.024)  |
| Mean dep. var.                              | 0.463                | 0.237                 | 0.237                  | 0.078                | 0.009                | 0.048                  | 0.494              | 0.222                 |
| Municipality FE                             | Yes                  | Yes                   | Yes                    | Yes                  | Yes                  | Yes                    | Yes                | Yes                   |
| Year FE                                     | Yes                  | Yes                   | Yes                    | Yes                  | Yes                  | Yes                    | Yes                | Yes                   |
| Observations                                | 2,121                | 2,121                 | 2,121                  | 2,121                | 2,121                | 2,121                  | 2,121              | 2,121                 |
| <b>Panel B: Household Amenities</b>         |                      |                       |                        |                      |                      |                        |                    |                       |
|                                             | Piped water          | Wood                  | Kerosene               | LPG                  | Modern fuel          | Traditional fuel       | Toilet (modern)    | Toilet (ordinary)     |
| FXShock $\times$ MigIntensity $\times$ Post | -0.0444<br>(0.0464)  | -0.1316**<br>(0.0624) | -0.0852***<br>(0.0321) | 0.1467**<br>(0.0621) | 0.1515**<br>(0.0634) | -0.1665***<br>(0.0618) | 0.0976<br>(0.0609) | -0.1950**<br>(0.0810) |
| Mean dep. var.                              | 0.567                | 0.756                 | 0.018                  | 0.112                | 0.113                | 0.880                  | 0.335              | 0.261                 |
| Municipality FE                             | Yes                  | Yes                   | Yes                    | Yes                  | Yes                  | Yes                    | Yes                | Yes                   |
| Year FE                                     | Yes                  | Yes                   | Yes                    | Yes                  | Yes                  | Yes                    | Yes                | Yes                   |
| Observations                                | 2,121                | 2,121                 | 2,121                  | 2,121                | 2,121                | 2,121                  | 2,121              | 2,121                 |
| <b>Panel C: Agricultural Practices</b>      |                      |                       |                        |                      |                      |                        |                    |                       |
|                                             | Dry season (Yes = 1) |                       |                        |                      | Wet season (Yes = 1) |                        |                    |                       |
|                                             | Self-cropped         | Share-cropped         | Rent-out               | Fallow               | Self-cropped         | Share-cropped          | Rent-out           | Fallow                |
| FXShock $\times$ MigIntensity $\times$ Post | -0.032***<br>(0.010) | -0.000<br>(0.003)     | 0.001<br>(0.003)       | 0.032***<br>(0.009)  | -0.007<br>(0.006)    | 0.002<br>(0.003)       | 0.002<br>(0.003)   | 0.004<br>(0.004)      |
| Mean dep. var.                              | 0.846                | 0.047                 | 0.021                  | 0.079                | 0.671                | 0.039                  | 0.018              | 0.264                 |
| Year FE                                     | Yes                  | Yes                   | Yes                    | Yes                  | Yes                  | Yes                    | Yes                | Yes                   |
| Household FE                                | Yes                  | Yes                   | Yes                    | Yes                  | Yes                  | Yes                    | Yes                | Yes                   |
| Observations                                | 24,689               | 24,689                | 24,689                 | 24,689               | 24,689               | 24,689                 | 24,689             | 24,689                |

*Notes:* This table reports the household-level effects of exposure to foreign exchange shocks as per equation (7). Panel A presents the effects on the share of households in the municipality population by ownership of assets, and Panel B reports the effects on households' access to amenities. Panel C presents the effects on households' agricultural cultivation practices in dry and wet seasons. All specifications include year fixed effects and either household or municipality fixed effects as indicated. Standard errors are clustered at the municipality level and are reported in parentheses. Significance: \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , and \* $p < 0.10$ .

as 40%. This evidence suggests some disengagement of households from farming activities, which may potentially lead to inefficient utilisation of agricultural land. Note that these patterns are more pronounced in the dry season, when irrigation constraints limit cultivation, than in the wet season. Moreover, the decline in self-cultivation and the rise in fallow land are consistent with labour withdrawal from agriculture, which we discuss in detail in the next section.

Table 3 provides further evidence of households' reduced engagement in agricultural activities. Farming households in municipalities of origin do not seem to invest in productivity-enhancing practices or technologies such as investing in infrastructure like canals for dry season cultivation. Rather they continue to rely on rainfed irrigation or traditional technologies. The use of key inputs such as seeds and hired labour also declines, while fertilizer and pesticide use shows no meaningful change. Crucially, households do not undertake capital investments in agriculture for equipment. These results suggest that although remittance inflows relax liquidity constraints, they do not translate into greater agricultural investment.

Table 3: Effect of *FXShock* on agricultural inputs and technology

| Panel A: Irrigation mode (Yes = 1)                           |                   |                    |                     |                   |                  |                   |                   |                      |                  |                   |
|--------------------------------------------------------------|-------------------|--------------------|---------------------|-------------------|------------------|-------------------|-------------------|----------------------|------------------|-------------------|
|                                                              | Wet season        |                    |                     |                   |                  | Dry season        |                   |                      |                  |                   |
|                                                              | Well              | Tubewell           | Canal               | Pond              | Rainfed          | Well              | Tubewell          | Canal                | Pond             | Rainfed           |
| MigShock $\times$ MigIntensity <sub>2001</sub> $\times$ Post | -0.001<br>(0.001) | 0.018**<br>(0.006) | -0.031**<br>(0.010) | 0.002*<br>(0.001) | 0.002<br>(0.009) | -0.000<br>(0.001) | 0.016*<br>(0.006) | -0.037***<br>(0.009) | 0.001<br>(0.001) | -0.013<br>(0.012) |
| Mean Dep. Var                                                | 0.002             | 0.087              | 0.123               | 0.002             | 0.452            | 0.002             | 0.084             | 0.196                | 0.003            | 0.557             |
| Observations                                                 | 24,689            | 24,689             | 24,689              | 24,689            | 24,689           | 24,689            | 24,689            | 24,689               | 24,689           | 24,689            |
| R <sup>2</sup>                                               | 0.242             | 0.590              | 0.383               | 0.245             | 0.428            | 0.237             | 0.583             | 0.383                | 0.238            | 0.399             |

| Panel B: Use of agricultural inputs (Yes = 1)                |                    |                  |                  |                   |                      |                     |                   |                   |                  |                     |
|--------------------------------------------------------------|--------------------|------------------|------------------|-------------------|----------------------|---------------------|-------------------|-------------------|------------------|---------------------|
|                                                              | Wet season         |                  |                  |                   |                      | Dry season          |                   |                   |                  |                     |
|                                                              | Seed               | Fertiliser       | Insecticide      | Equipment hire    | Labour hire          | Seed                | Fertiliser        | Insecticide       | Equipment hire   | Labour hire         |
| MigShock $\times$ MigIntensity <sub>2001</sub> $\times$ Post | -0.021*<br>(0.011) | 0.002<br>(0.011) | 0.012<br>(0.008) | -0.004<br>(0.014) | -0.028***<br>(0.009) | -0.028**<br>(0.013) | -0.011<br>(0.009) | 0.017*<br>(0.009) | 0.014<br>(0.013) | -0.018**<br>(0.008) |
| Mean Dep. Var                                                | 0.250              | 0.524            | 0.198            | 0.215             | 0.390                | 0.260               | 0.382             | 0.158             | 0.189            | 0.250               |
| Observations                                                 | 15,922             | 15,922           | 15,922           | 15,922            | 15,922               | 15,922              | 15,922            | 15,922            | 15,922           | 15,922              |
| R <sup>2</sup>                                               | 0.497              | 0.652            | 0.575            | 0.542             | 0.570                | 0.457               | 0.628             | 0.529             | 0.557            | 0.502               |

Notes: This table reports the household-level effects of exposure to foreign exchange shocks as per equation (7). The unit of observations is household-by-year for 2016, 2017, 2018. Panel A shows the effect on various irrigation technologies (with each being a binary variable) for wet and dry seasons. Panel B reports the use of various agricultural inputs (each being binary variable). Standard errors clustered at the municipality level are presented in parentheses. Significance: \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

Overall, the evidence indicates that favourable foreign exchange shocks improve household welfare but fail to spur productive transformation at the local level, even after two decades. Remittances ease subsistence constraints but these gains do not result in efficient land use or investment in agricultural capital. The outcome is improved living standards without a strengthened productive base.

## 5.2 Sectoral Reallocation of Labour

This section examines how *FXShock* influences municipal-level labour market dynamics. Higher household purchasing power from favorable exchange rate movements may lead to greater investment in education and skills development. Such human capital accumulation may facilitate transitions from the low-return subsistence agriculture into higher-wage employment or individuals may pursue into more specialised and skill-intensive occupations. Table 4 presents the estimated effects on worker reallocation across sectors and industries, shifts in occupational structure, and changes in the nature of employment.

Panel A shows that municipalities more exposed to favourable *FXShock* experienced a six percentage point decline in agricultural employment between 2001 and 2021. This withdrawal from agriculture is mirrored by a seven percentage point increase in employment in the services sector, while employment in industry shows no significant change. These results suggest that remittance-induced income gains accelerate structural transformation in origin areas by facilitating labour exit from low-productivity farming. Such



Table 4: Impact of *FXShock* on employment reallocation and occupation

|                                              | Panel A: By Broad Sectors |                  |                     | Panel B: By Selected Industry |                  |                    |                   |                     |                     |
|----------------------------------------------|---------------------------|------------------|---------------------|-------------------------------|------------------|--------------------|-------------------|---------------------|---------------------|
|                                              | Agri<br>(1)               | Industry<br>(2)  | Service<br>(3)      | Manuf.<br>(4)                 | Const.<br>(5)    | Hospitality<br>(6) | Retail<br>(7)     | Public Admin<br>(8) | Others<br>(9)       |
| MigShock $\times$ MigIntensity $\times$ Post | -0.058**<br>(0.025)       | 0.001<br>(0.014) | 0.069***<br>(0.019) | -0.003<br>(0.009)             | 0.002<br>(0.010) | 0.016**<br>(0.006) | 0.014*<br>(0.008) | -0.041**<br>(0.017) | 0.113***<br>(0.033) |
| Mean Dep. Var.                               | 0.447                     | 0.129            | 0.424               | 0.045                         | 0.038            | 0.010              | 0.005             | 0.010               | 0.070               |
| Municipality FE                              | Yes                       | Yes              | Yes                 | Yes                           | Yes              | Yes                | Yes               | Yes                 | Yes                 |
| Year FE                                      | Yes                       | Yes              | Yes                 | Yes                           | Yes              | Yes                | Yes               | Yes                 | Yes                 |
| Observations                                 | 2,121                     | 2,121            | 2,121               | 2,121                         | 2,121            | 2,121              | 2,121             | 2,121               | 2,121               |
| R <sup>2</sup>                               | 0.904                     | 0.807            | 0.893               | 0.805                         | 0.760            | 0.894              | 0.863             | 0.590               | 0.713               |

|                                              | Panel C: By Occupation |                     |                      |                    |                    | Panel D: By Employment Type |                    |                      |                      |
|----------------------------------------------|------------------------|---------------------|----------------------|--------------------|--------------------|-----------------------------|--------------------|----------------------|----------------------|
|                                              | Agriculture<br>(1)     | Managers<br>(2)     | Professionals<br>(3) | Elementary<br>(4)  | Technicians<br>(5) | Employees<br>(6)            | Employers<br>(7)   | Self-employed<br>(8) | Unpaid family<br>(9) |
| MigShock $\times$ MigIntensity $\times$ Post | -0.097***<br>(0.026)   | 0.054***<br>(0.009) | -0.008<br>(0.010)    | 0.039**<br>(0.015) | -0.004<br>(0.009)  | 0.071***<br>(0.017)         | 0.013**<br>(0.006) | 0.039<br>(0.028)     | -0.107**<br>(0.045)  |
| Mean Dep. Var.                               | 0.666                  | 0.017               | 0.023                | 0.128              | 0.012              | 0.154                       | 0.009              | 0.162                | 0.228                |
| Municipality FE                              | Yes                    | Yes                 | Yes                  | Yes                | Yes                | Yes                         | Yes                | Yes                  | Yes                  |
| Year FE                                      | Yes                    | Yes                 | Yes                  | Yes                | Yes                | Yes                         | Yes                | Yes                  | Yes                  |
| Observations                                 | 2,121                  | 2,121               | 2,121                | 2,121              | 2,121              | 2,121                       | 2,121              | 2,121                | 2,121                |
| R <sup>2</sup>                               | 0.934                  | 0.799               | 0.842                | 0.883              | 0.861              | 0.936                       | 0.539              | 0.900                | 0.673                |

Notes: This table reports the labour market effects of exposure to foreign exchange shocks as per equation (7). The unit of observation is municipality-year for census rounds 2001, 2011, and 2021. *Post* is an indicator for the baseline year 2001, taking the value 0 in 2001 and 1 in subsequent years. Dependent variables are municipal employment shares (in proportions) by sector, industry, occupation, and employment type. Panel A: *Agriculture* includes agriculture, forestry, and fishing; *Industry* includes manufacturing, construction, and water and sewage; *Services* aggregates the remaining sectors. Panel B disaggregates services into subcategories: *Hospitality, Wholesale and Retail Trade, Public Administration*, and *Others* (which includes finance, education, and remaining sectors). Panel C presents occupational categories as recorded in the censuses. Panel D shows employment types. All regressions include municipality and year fixed effects. Standard errors are clustered at the municipality level and reported in parentheses. Significance: \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

labour reallocation aligns with household-level results discussed in section 5.1, which shows reduced households engagement in farming activities.

Panel B observes that the expansion of employment is concentrated in a few service industries. The largest gains occur in retail trade, where employment is more than tripled from its initial level, and in the hospitality sector where employment increases by roughly 60%. Employment in other services like education and financial sector also rises by more than 150%. By contrast, manufacturing and construction display no meaningful change from their baseline levels, while the share of workforce in public administration experiences a sharp decline. Overall, these results indicate that *FXShock* stimulates employment growth almost exclusively in non-tradable services rather than in industrial sectors.

Panel C reports how the changed occupational structure of the labour market due to *FXShock*. In migrant-intensive municipalities, the share of agricultural-related occupations declines by 15%. Managerial and elementary occupations see the largest gains, which reflects growing demand for both coordination roles and entry-level service jobs. By contrast, professional and technical occupations show no systematic change. Overall, remittance-induced demand appears to increase employment in low- and mid-skill ser-

vices but failed to create high-skill professional positions.

Panel D shows changes in composition of employment in high-migration municipalities. The share of unpaid family workers declines by about 47% from its baseline level with equivalent rise in the share of wage employees. The proportion of employers increases by around 144%, although from a small base. Self-employment shows no significant change. These results indicate a clear reallocation of labour toward paid, non-farm and employer-based work and away from unpaid family labour in agriculture.

In summary, the results indicate that *FXShock* triggers significant labour reallocation toward non-tradable services out of agriculture. The change also includes modest upgrading of occupations and a shift away from unpaid family workers. Overall, while migration exposure reduces agricultural dependence, it has not facilitated a transition toward high-quality employment, which may throw some light on Nepal's persistent underdevelopment over the past two decades.

### 5.3 Firm Level Outcomes

This section examines the entrepreneurial responses to *FXShock*, which may occur through multiple channels. Rising remittance income may expand local demand for goods and services or enable households to invest in non-farm enterprises. Returning migrants bring new skills, organisational practices, or risk-taking capacity that could foster business creation in origin areas. These dynamics collectively support the emergence and expansion of new firms and stimulate local economic transformation.

Table 5 reports that areas experiencing favourable currency shocks are associated with a higher number of firms. Panel A shows that enterprises of all sizes benefit from remittance inflows, with effects particularly pronounced among medium and larger firms. Panel B extends these findings to employment: municipalities that are more exposed to favourable currency shocks exhibit significantly greater employment. Together, these results suggest remittances facilitate presence of business firms and employment, thus, enhancing local economic dynamism.

Figure 3 disaggregates firm presence by industrial sector, and reveals that the effects of *FXShock* are highly uneven across industries. The strongest and most statisti-

Table 5: Effect of *FXShock* on the number of firms and employment

|                                                    | Total               | Micro            | Small               | Medium             | Large               |
|----------------------------------------------------|---------------------|------------------|---------------------|--------------------|---------------------|
| <b>Panel A: Number of firms (Log)</b>              |                     |                  |                     |                    |                     |
| $\text{FXShock} \times \text{MigIntensity}_{2001}$ | 0.065***<br>(0.025) | 0.043<br>(0.031) | 0.095***<br>(0.026) | 0.060**<br>(0.029) | 0.107***<br>(0.030) |
| Mean dep. var.                                     | 6.61                | 5.62             | 6.01                | 3.29               | 0.84                |
| Observations                                       | 707                 | 707              | 707                 | 707                | 707                 |
| R <sup>2</sup>                                     | 0.476               | 0.533            | 0.437               | 0.438              | 0.351               |
| <b>Panel B: Number of person engaged (Log)</b>     |                     |                  |                     |                    |                     |
|                                                    | Total               | Micro            | Small               | Medium             | Large               |
| $\text{FXShock} \times \text{MigIntensity}_{2001}$ | 0.087***<br>(0.026) | 0.043<br>(0.031) | 0.087***<br>(0.026) | 0.063*<br>(0.033)  | 0.237**<br>(0.101)  |
| Mean dep. var.                                     | 7.60                | 5.54             | 6.99                | 5.94               | 2.84                |
| Observations                                       | 707                 | 707              | 707                 | 707                | 707                 |
| R <sup>2</sup>                                     | 0.434               | 0.533            | 0.419               | 0.436              | 0.309               |

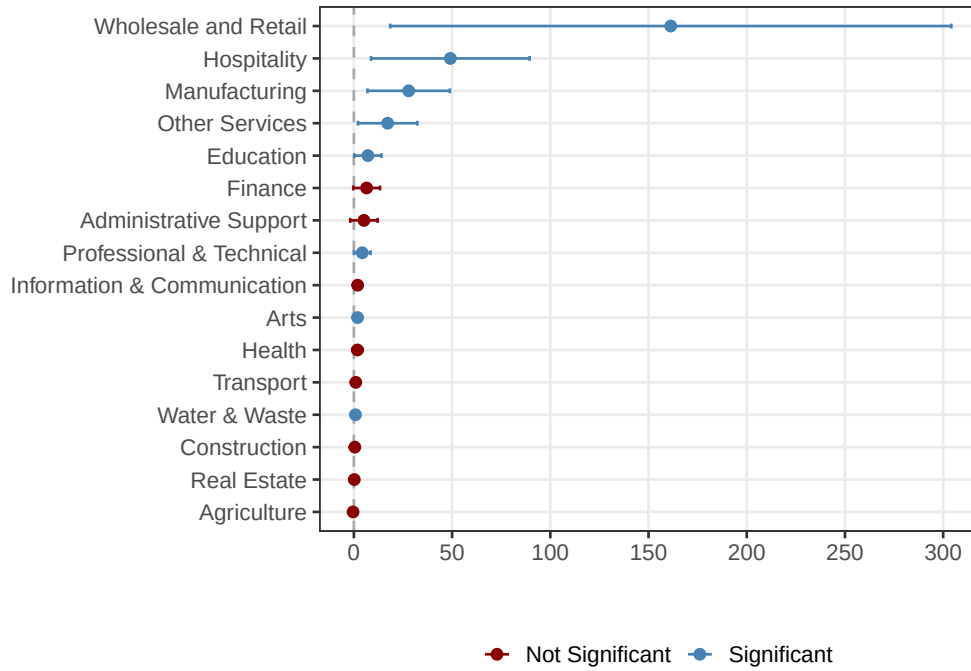
*Notes:* The table presents estimates of the effect of municipality-level exposure to migration-induced foreign exchange shocks on firm presence, based on a cross-sectional specification of equation (7) using data from the 2018 Economic Census. Panel A presents effects on the number of registered establishments, while Panel B shows effects on total employment within establishments, both expressed in logarithmic form using  $\log(1 + y)$ . Firm size categories follow the Central Bureau of Statistics classification: Micro (1 worker), Small (2–9), Medium (10–50), and Large (51+). All regressions include district fixed effects. Standard errors are clustered at the municipality level and reported in parentheses. Significance: \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

cally significant effects are observed in wholesale and retail trade, hospitality, manufacturing, and education, indicating that remittance windfalls primarily stimulate demand in consumption-oriented and locally traded sectors. By contrast, construction, transport, information and communication, and professional services show no meaningful response, while agriculture-based firms slightly decline. This pattern indicates demand-side forces rather than investment in tradable or capital-intensive activities as the key facilitator of firms' presence.

## 6 Conclusion

This essay examines the causal effects of remittance windfalls, facilitated through foreign exchange fluctuations, on household welfare, labour markets, and firm dynamics in Nepal. It employs a shift-share instrumental variable design that leverages plausibly exogenous

Figure 3: Number of establishment by major industrial sectors



*Notes:* The figure shows estimated effects of remittance gain due to foreign exchange shocks on the number of enterprises across major industrial sectors. Points represent cross-section coefficient estimates based on equation (7). Horizontal lines denote the 95% confidence intervals and color markers indicate statistically significant effects at the 5% level.

variation in destination-country currency movements interacted with historical migration patterns at the municipality level. The analysis first establishes that migrants' destination choices and the value of remittance inflows respond systematically to relative exchange rate shocks. Then, it goes on to estimate the downstream effects on households, local labour markets, and firms.

Favorable foreign exchange shocks that raise the remittance income produce clear welfare gains as households accumulate more assets and gain access to modern amenities. However, they simultaneously reduce household engagement in agriculture by decreasing the share of households with self-cultivation and land fallow, and cutting back on key input use. Consequently, labour reallocates into low-productivity services away from agriculture. Local areas witness the expansion of non-agriculture enterprises, most likely driven by remittance-financed household demand.

Overall, migration seems to enhance household welfare but does not automatically

induce productive transformation of local economies. The evidence suggests that remittances primarily serve as a consumption-smoothing mechanism rather than a catalyst for structural change. These results underscore Nepal's persistent reliance on external labour markets over the past two decades, and indicates that the prospects for sustained productivity growth within the local economy through this channel remains limited.

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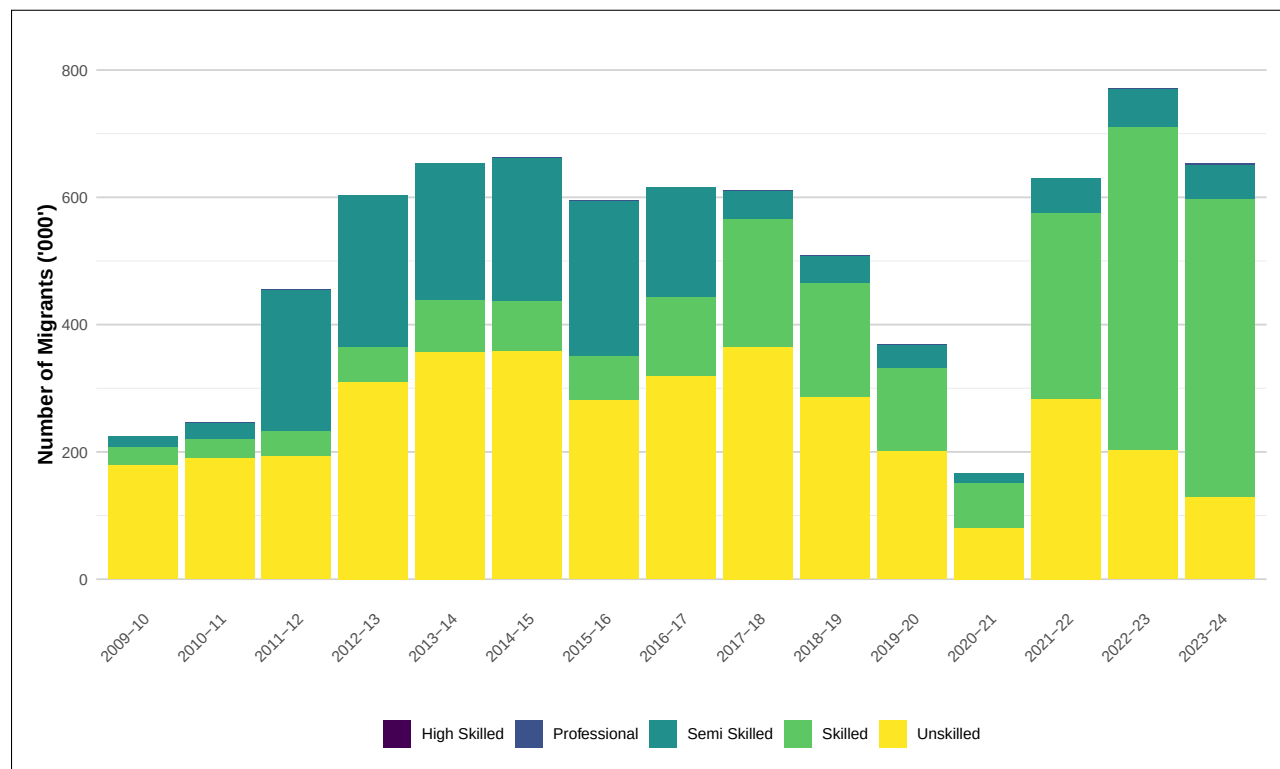
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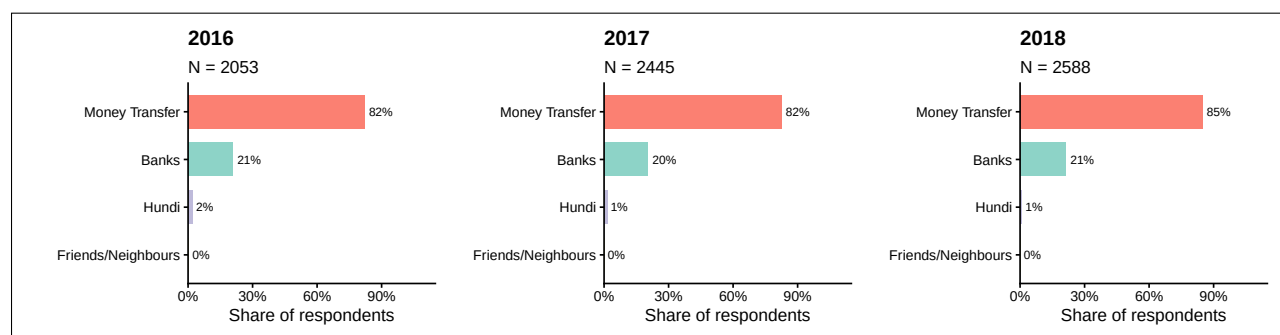
## A Additional Tables and Figures

Figure A1: Labour approval by skill-types



Notes: The figure shows the distribution of migrants outflow by skill-types.

Figure A2: Methods of Sending Remittances Across Survey Waves (2016–2018)



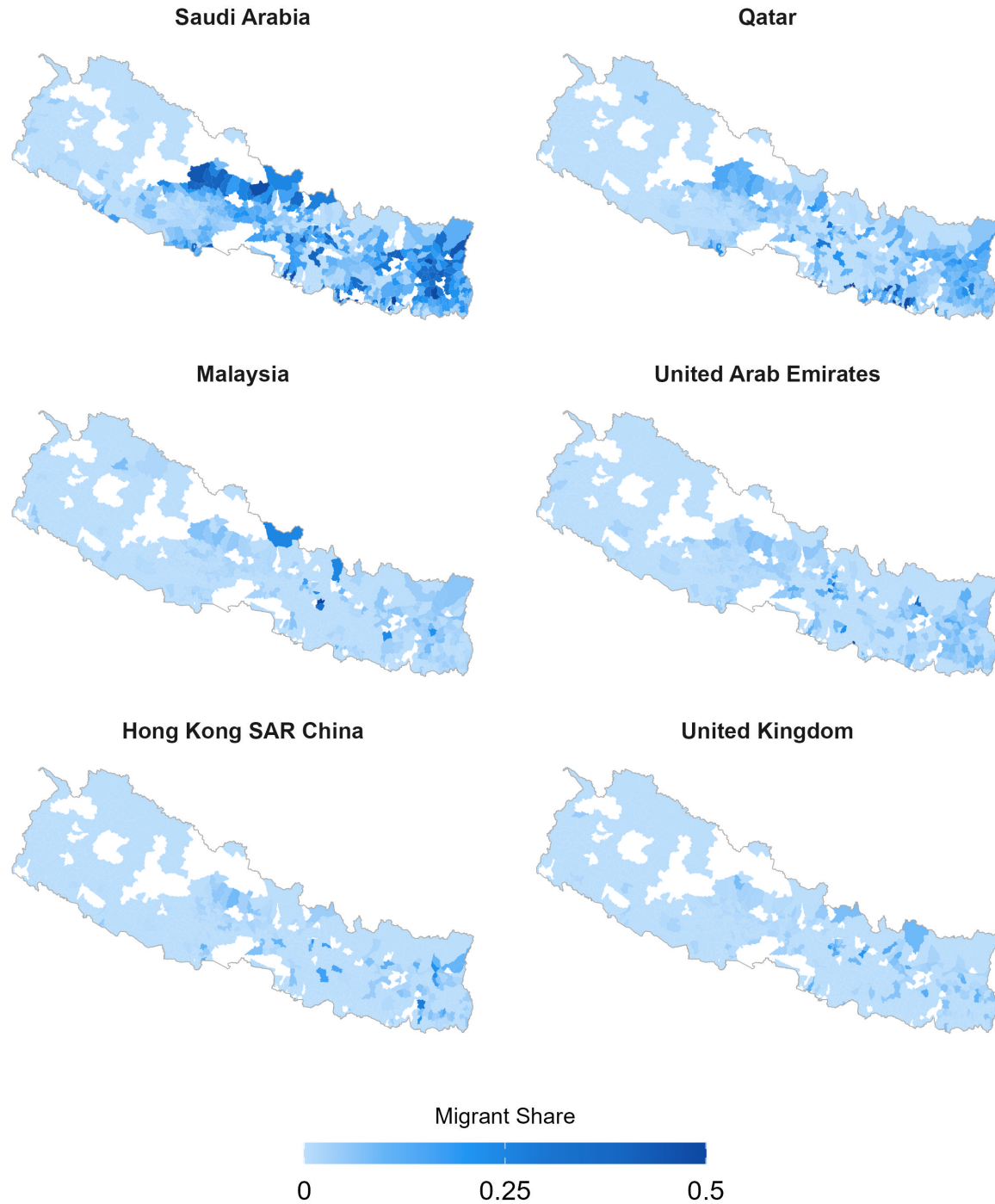
Notes: The figure shows the distribution of remittance transfer methods used by Nepali migrant-sending households across the 2016–2018 waves of Nepal Household Risk and Vulnerability Survey (HRVS).

Table A1: Migrant share in top-20 destination countries (2001)

| Country                 | Migrant Share |
|-------------------------|---------------|
| Saudi Arabia            | 0.445         |
| Qatar                   | 0.161         |
| United Arab Emirates    | 0.085         |
| China (incl. Hong Kong) | 0.079         |
| Malaysia                | 0.047         |
| United Kingdom          | 0.038         |
| Kuwait                  | 0.027         |
| United States           | 0.022         |
| Bahrain                 | 0.021         |
| Japan                   | 0.018         |
| South Korea             | 0.018         |
| Singapore               | 0.016         |
| Germany                 | 0.007         |
| Australia               | 0.005         |
| Bhutan                  | 0.003         |
| Bangladesh              | 0.002         |
| Maldives                | 0.002         |
| Pakistan                | 0.002         |
| France                  | 0.001         |
| Russia                  | 0.001         |
| Other Countries         | 0.000         |

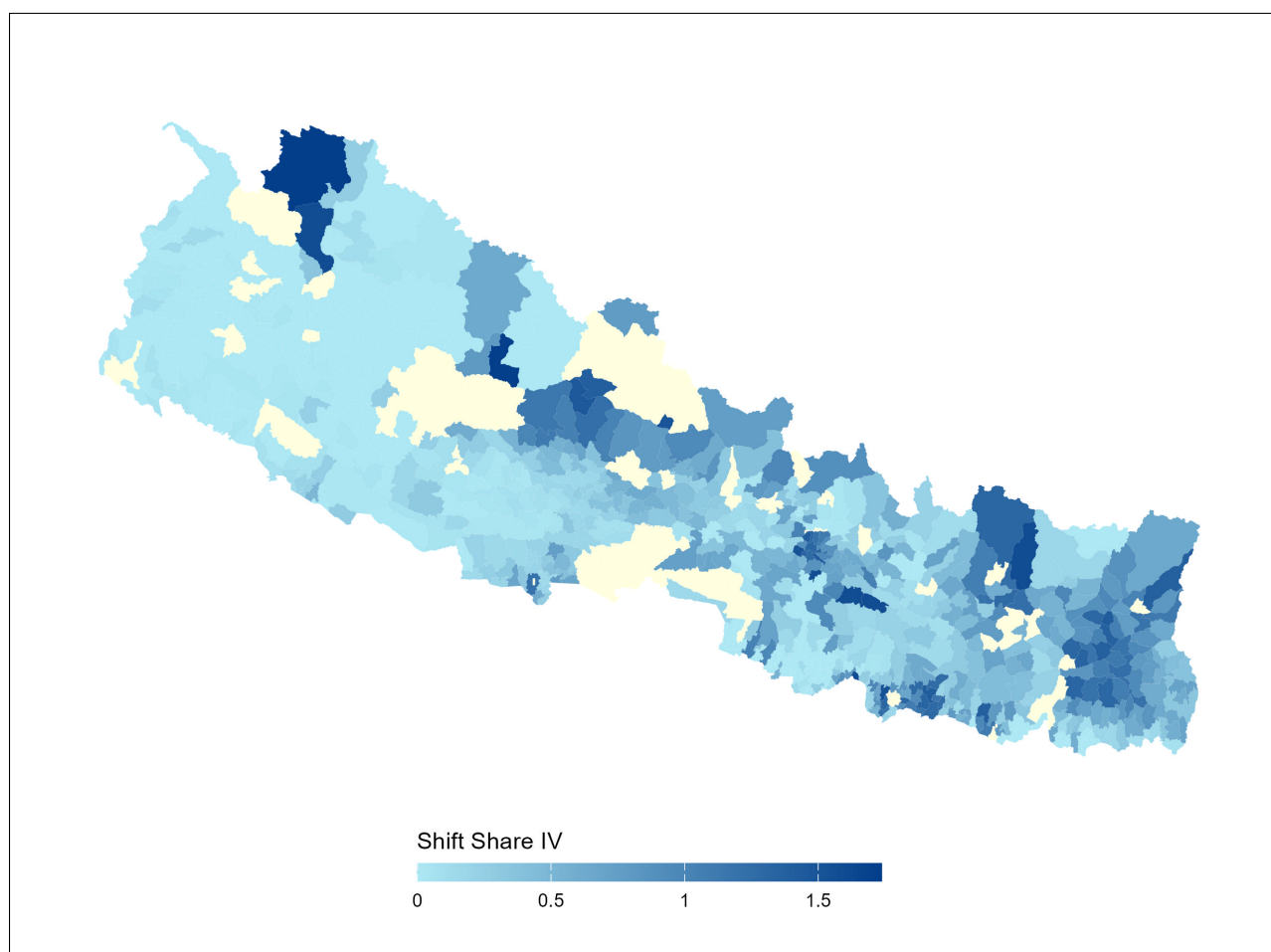
The table showcase share of migrants in top-20 destination countries, other than India, in 2001.

Figure A3: Municipality-Level Migration Exposure in 2001



*Notes:* The figure maps municipality-level migration exposure, measured as the share of migrants from each municipality employed in different destination countries in the baseline year 2001. Darker shades indicate higher exposure, reflecting areas with a greater concentration of migrants in foreign employment relative to total out-migrants.

Figure A4: Distribution of Shift-Share Instrument index



*Notes:* The figure shows the shift-share distribution across municipalities.

Figure A5: Distribution of Baseline Migrant Intensity across Municipalities in 2001

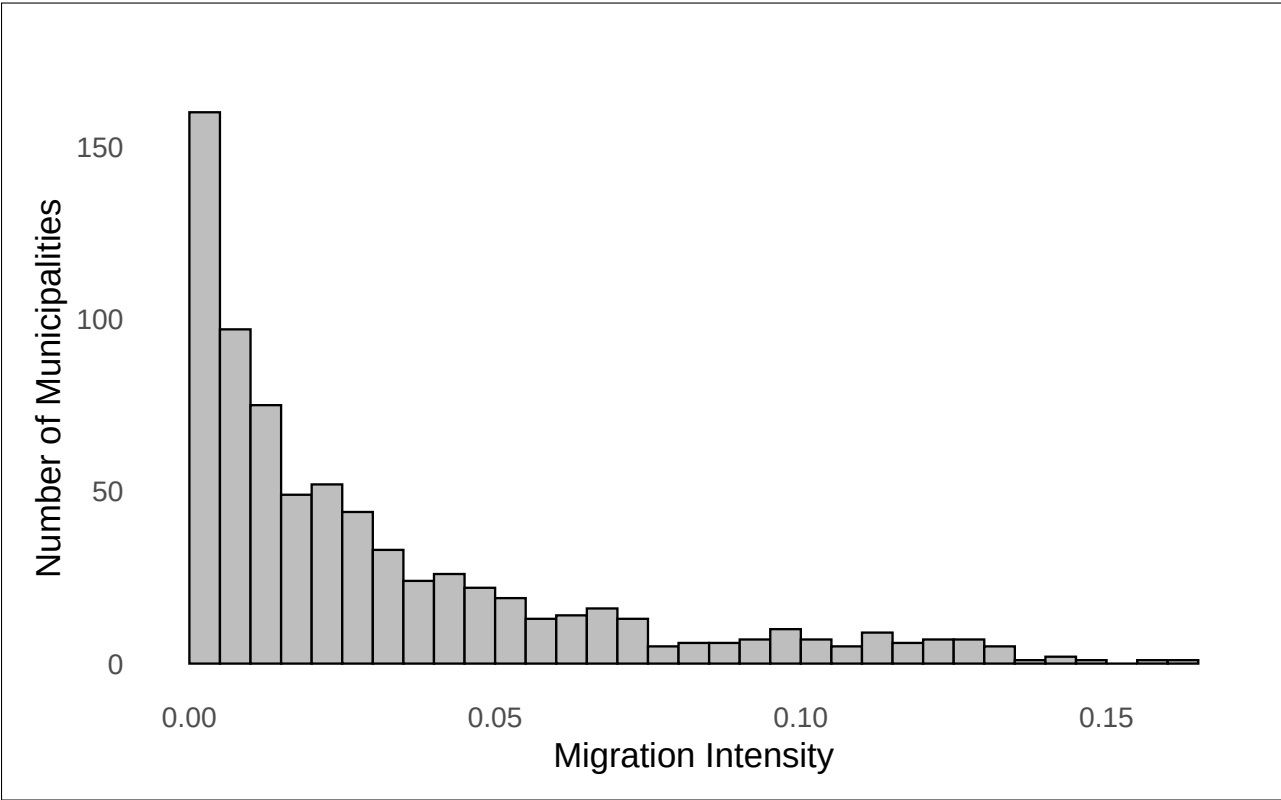


Table A2: Summary Statistics

| Variable                         | 2001 (N = 744) |       |       |       | 2011 (N = 753) |       |       |       | 2021 (N = 753) |       |       |       |
|----------------------------------|----------------|-------|-------|-------|----------------|-------|-------|-------|----------------|-------|-------|-------|
|                                  | Mean           | SD    | Min   | Max   | Mean           | SD    | Min   | Max   | Mean           | SD    | Min   | Max   |
| Forex Shock                      | 0.962          | 0.020 | 0.948 | 1.101 | 1.012          | 0.072 | 0.711 | 1.734 | 1.552          | 0.122 | 0.604 | 2.101 |
| <b>Sectoral Employment</b>       |                |       |       |       |                |       |       |       |                |       |       |       |
| Primary                          | 0.769          | 0.147 | 0.090 | 1.000 | 0.738          | 0.157 | 0.024 | 0.975 | 0.676          | 0.177 | 0.051 | 0.928 |
| Secondary                        | 0.101          | 0.064 | 0.003 | 0.458 | 0.071          | 0.053 | 0.001 | 0.379 | 0.101          | 0.069 | 0.008 | 0.384 |
| Tertiary                         | 0.130          | 0.099 | 0.006 | 0.674 | 0.191          | 0.118 | 0.024 | 0.786 | 0.223          | 0.125 | 0.062 | 0.811 |
| <b>Industry-level Employment</b> |                |       |       |       |                |       |       |       |                |       |       |       |
| Agriculture                      | 0.768          | 0.148 | 0.089 | 1.000 | 0.736          | 0.158 | 0.021 | 0.975 | 0.674          | 0.178 | 0.045 | 0.928 |
| Manufacturing                    | 0.067          | 0.047 | 0.003 | 0.422 | 0.042          | 0.037 | 0.001 | 0.328 | 0.029          | 0.028 | 0.001 | 0.255 |
| Construction                     | 0.023          | 0.023 | 0.000 | 0.163 | 0.028          | 0.019 | 0.001 | 0.105 | 0.068          | 0.047 | 0.002 | 0.275 |
| Wholesale-Retail                 | 0.052          | 0.040 | 0.003 | 0.246 | 0.048          | 0.039 | 0.002 | 0.258 | 0.087          | 0.050 | 0.012 | 0.261 |
| Public Service/Defence           | 0.017          | 0.023 | 0.001 | 0.204 | 0.008          | 0.011 | 0.000 | 0.125 | 0.003          | 0.009 | 0.000 | 0.175 |
| <b>Employment by Occupation</b>  |                |       |       |       |                |       |       |       |                |       |       |       |
| Professionals                    | 0.019          | 0.013 | 0.001 | 0.116 | 0.036          | 0.018 | 0.004 | 0.156 | 0.035          | 0.016 | 0.004 | 0.133 |
| Managers                         | 0.004          | 0.006 | 0.000 | 0.052 | 0.010          | 0.015 | 0.000 | 0.184 | 0.039          | 0.030 | 0.004 | 0.212 |
| Assistants                       | 0.013          | 0.016 | 0.001 | 0.166 | 0.010          | 0.010 | 0.001 | 0.107 | 0.009          | 0.009 | 0.001 | 0.072 |
| Technicians                      | 0.009          | 0.014 | 0.000 | 0.169 | 0.014          | 0.016 | 0.001 | 0.099 | 0.013          | 0.012 | 0.001 | 0.095 |
| Elementary                       | 0.126          | 0.098 | 0.003 | 0.484 | 0.097          | 0.081 | 0.002 | 0.394 | 0.167          | 0.114 | 0.007 | 0.544 |
| <b>Employment Status</b>         |                |       |       |       |                |       |       |       |                |       |       |       |
| Employee                         | 0.191          | 0.140 | 0.018 | 0.679 | 0.229          | 0.134 | 0.021 | 0.629 | 0.239          | 0.138 | 0.051 | 0.845 |
| Employer                         | 0.036          | 0.031 | 0.002 | 0.368 | 0.019          | 0.013 | 0.001 | 0.092 | 0.011          | 0.008 | 0.001 | 0.077 |
| Self-employed                    | 0.710          | 0.150 | 0.201 | 0.966 | 0.730          | 0.140 | 0.317 | 0.958 | 0.632          | 0.141 | 0.133 | 0.891 |
| Unpaid Family Worker             | 0.063          | 0.043 | 0.004 | 0.620 | 0.022          | 0.033 | 0.000 | 0.453 | 0.118          | 0.049 | 0.008 | 0.586 |

Table A3: Effect of Migration Shock on Types of Employment

|                                                | Employment outcomes (indicator = 1) |                   |                  |                    |                  |
|------------------------------------------------|-------------------------------------|-------------------|------------------|--------------------|------------------|
|                                                | All                                 | Wage-based        |                  | Self-employed      |                  |
|                                                | Worked                              | Agri              | Non-agri         | Agri               | Non-agri         |
| MigShock $\times$ MigIntensity <sub>2001</sub> | 0.007*<br>(0.005)                   | 0.004*<br>(0.003) | 0.003<br>(0.004) | -0.007*<br>(0.004) | 0.000<br>(0.002) |
| Year FE                                        | Yes                                 | Yes               | Yes              | Yes                | Yes              |
| Individual FE                                  | Yes                                 | Yes               | Yes              | Yes                | Yes              |
| Mean Dep. Var                                  | 0.518                               | 0.045             | 0.202            | 0.668              | 0.085            |
| Observations                                   | 50,446                              | 37,658            | 37,658           | 37,658             | 37,658           |
| R <sup>2</sup>                                 | 0.813                               | 0.502             | 0.587            | 0.566              | 0.595            |
| Within R <sup>2</sup>                          | 0.000                               | 0.000             | 0.000            | 0.000              | 0.000            |

The standard error is clustered at the municipality level. Significance levels: \*\*\* 0.01, \*\* 0.05, \* 0.1.